MA'ADEN معادن

MA'ADEN ENGINEERING MANUAL

Storm Water Drainage And Gravity Sewers

MD-101-SMEM-EG-CV-SPC-0004

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1. INTRODUCTION

1.1 Scope

This document specifies the minimum requirements for the construction of storm water drainage and gravity sewers for all Ma'aden projects.

This document is intended to be used in conjunction with the applicable documents listed herein.

1.2 Purpose

The purpose of this specification is to:

- Identify applicable Standards and guidelines to be used for the Work
- Set the minimum engineering standard for storm water drainage and gravity sewers
- Define the general requirements for the Work.

2. KEY TERMS AND DEFINITIONS

Contractor	An entity, engaged or approved by the Owner, entrusted with a contract to perform Work or cause Work to be performed.
CPP	Corrugated Polyethylene Pipe
Employer	Saudi Arabian Mining Company (Ma'aden) and its authorized representatives and respective affiliates.
HDPE	High Density Polyethylene.
ITP	Inspection and Test Plan.
NPS	Nominal Pipe Size
PVC	Poly Vinyl Chloride
SDR	Standard Dimension Ratio
Site	Geographical location where Services are to be performed.
VCP	Vitrified Clay Pipe
Work	The entire work and Services and all things which the Contractor is or may be required to execute, supply and do under the Contract having regard to the Scope of Project, or to cause to be executed, supplied or done, for the purposes of constructing and Commissioning the Plant.

3. REFERENCED DOCUMENTS

This specification requires compliance with the provisions of the latest revision of the relevant standards or codes of the following authorities and technical organizations, as applicable, and specifications which shall be considered to form part of this specification, where applicable.

3.1 Statutory Regulations

All Work performed and materials furnished shall comply with the applicable statutory regulations, codes and other requirements of the Kingdom of Saudi Arabia (including but not limited to directives of the High Commission for Industrial Security, Royal Commission regulations applicable to the Site, regulations of the Presidency of Meteorology and Environment, Saudi Building Code and other applicable codes and regulations). The Contractor shall ensure that he is fully aware and informed of all Government laws, local ordinances, regulations and all orders and decrees of bodies or tribunals having any jurisdiction or authority over the Project and/or Site, which in any manner affect the Project and those engaged or employed on the Project, or which in any way affect the conduct of the Work. The Contractor shall at all times observe and comply with all such Government and/or Ministry laws, bylaws, ordinances, codes, regulations, orders and decrees.

3.2 Industry Codes and Standards

All equipment and systems shall conform to the following.

3.2.1 Local Codes and Standards

- Royal Commission Guidelines.
- Engineering Manual.
- Environmental Regulation Conservation Document (ERCD).

3.2.2 International Codes and Standards

3.2.2.1 ASTM – American Society for Testing and Materials

ASTM A36	Standard Specification for Carbon Structural Steel
ASTM A53	Standard Specification for Pipe, Steel, Black and Hot dipped, Zinc-coated, Welded and Seamless
ASTM A74	Standard Specification for Cast Iron Soil Pipe and Fittings
ASTM A90/A90M	Standard Test Method for Weight (Mass) of Coating on Iron and Steel Articles with Zinc or Zinc-Alloy Coatings
ASTM A126	Standard Specification for Gray Iron Castings for Valves, Flanges, and Pipe Fittings
ASTM A234	Standard Specification for Piping Fittings of Wrought Carbon Steel and Alloy Steel for Moderate and Elevated Temperatures
ASTM A672	Standard Specification for Electric-Fusion-Welded Steel pipe for High Pressure Service at Moderate Temperatures
ASTM A760/A760M	Standard Specification for Corrugated Steel Pipe, Metallic-Coated for Sewers and Drains
ASTM A796/A796M	Standard Practice for Structural Design of Corrugated Steel Pipe, Pipe-Arches, and Arches for Storm and Sanitary Sewers and Other Buried Applications

ASTM C14	Standard Specification for Concrete Sewer, Storm Drain and Culvert Pipe
ASTM C33	Standard Specification for Concrete Aggregates
ASTM C76	Standard Specification for Reinforced Concrete Culvert, Storm Drain and Sewer Pipe
ASTM C425	Standard Specification for Compression Joints for Vitrified Clay Pipe and Fittings
ASTM C443	Standard Specification for Joints for Circular Concrete Sewer and Culvert Pipe, Using Rubber Gaskets
ASTM C478	Standard Specification for Precast Reinforced Concrete Manhole Sections
ASTM C564	Standard Specification for Rubber Gaskets for Cast Iron Soil Pipe and Fittings
ASTM C700	Standard Specification for Vitrified Clay Pipe, Extra Strength, Standard Strength and Perforated
ASTM C969	Standard Practice for Infiltration and Exfiltration Acceptance Testing of Installed Precast Concrete Pipe Sewer Lines
ASTM C1417	Standard Specification for Manufacture of Reinforced Concrete Sewer, Storm Drain, and Culvert Pipe for Direct Design
ASTM C1433	Standard Specification for Precast Reinforced Concrete Box Sections for Culverts, Storm Drains and Sewers
ASTM D422	Standard Test Method for Particle – Size Analysis of Soils
ASTM D698	Standard Test Methods for Laboratory Compaction Characteristics of Soil Using Standard Effort (12 400 ft-lbf/ft^3 (600 kN-m/m^3))
ASTM D1248	Standard Specification for Polyethylene Plastics Extrusion Materials for Wire and Cable
ASTM D1785	Standard Specification for Poly(Vinyl Chloride) (PVC) Plastic Pipe, Schedules 40, 80 and 120
ASTM D2321	Standard Practice for Underground Installation of Thermoplastic Pipe for Sewers and Other Gravity Flow Applications
ASTM D2564	Standard Specification for Solvent Cements for Poly(Vinyl Chloride) (PVC) Plastic Piping Systems
ASTM D2855	Standard Practice for Making Solvent-Cemented Joints with Poly(Vinyl Chloride) (PVC) Pipe and Fittings
ASTM D2996	Standard Specification for Filament-Wound 'Fiberglass' (Glass-Fiber-Reinforced Thermosetting-Resin) Pipe

ASTM D2997	Standard Specification for Centrifugally Cast 'Fiberglass' (Glass-Fiber-Reinforced Thermosetting-Resin) Pipe
ASTM D3212	Standard Specification for Joints for Drain and Sewer Plastic Pipes Using Flexible Elastomeric Seals
ASTM D3350	Standard Specification for Polyethylene Plastics Pipe and Fittings Materials
ASTM D3262	Standard Specification for 'Fiberglass' (Glass-Fiber-Reinforced Thermosetting-Resin) Sewer Pipe
ASTM F477	Standard Specification for Elastomeric Seals (Gaskets) for Joining Plastic Pipe

3.2.2.2 AASHTO – American Association of State Highway and Transportation Officials

AASHTO M 36	Standard Specification for Corrugated Steel Pipe, Metallic Coated, for Sewers and Drains
AASHTO M 294	Standard Specification for Corrugated Polyethylene Pipe, 300 – to 1500 mm Diameter

3.2.2.3 ISO – International Organization for Standardization

ISO 2531	Ductile iron pipes, fittings, accessories and their joints for water applications

3.2.3 Alternative Codes

Alternative codes may be used for storm water drainage and gravity sewers if approved by the Employer.

3.3 Ma'aden Standard Specifications

MD-101-SMEM-EG-CV-SPC-0001	Earthworks
MD-101-SMEM-EG-ST-SPC-0005	Concrete Works
MD-101-SMPM-PM-QA-PEI-0001	Contractor Quality Requirements

4. MATERIALS

4.1 General

All materials supplied shall comply with the applicable Standards as specified above and meet the approval of the Employer.

Where pipes or component materials are not covered by the specified Standards, complete details and manufacturer's specifications shall be submitted to the Employer for approval prior to use on Site.

4.2 Reticulation Materials

The following materials shall be selected unless noted otherwise on the drawings.

4.2.1 Poly Vinyl Chloride (PVC) Pipes and Fittings

PVC pipes DN300 or less shall be minimum Schedule 40 pipe in accordance with ASTM D1785.

The jointing materials to be used with PVC piping shall be solvent cement in accordance with ASTM D2855, using cement in accordance with ASTM D2564 or shall have elastomeric ring gaskets in accordance with ASTM D3212.

4.2.2 High-Density Polyethylene (HDPE) Pipe

HDPE pipe and fittings shall be manufactured from virgin high-density polyethylene resin, in accordance with ASTM D3350, Cell Class PE345434C minimum, with a standard dimension ratio (SDR) as specified on the drawings.

Joints shall be butt fusion welded, socket fusion welded or flanged.

Alternate joining method shall consist of arch-band split sleeve couplings made of carbon steel conforming to ASTM A36 or stainless steel grade 304 L or 316 L. The coupling shall be designed with closure plates and sealing pads for installation and a leak-proof seal. O-Ring gaskets shall be used and shall consist of ethylene propylene diene monomer (EPDM), nitrile, silicon or polyisoprene as required for the service intended. Couplings shall allow for angular deflection, expansion and vibration if required. If a restrained joint is required, it shall be accomplished with restraining rings welded to the exterior of the pipe and a restraining-type coupling.

HDPE mechanical coupling joints are also acceptable if approved by the Employer.

Unless otherwise shown on the drawings, wall thickness shall be specified as follows:

Table 4-1

NPS (Nominal Pipe Size) (mm)	SDR (Standard Dimension Ratio)
100 to 400	17
450 to 1200	32.5

4.2.3 Concrete Sewer Pipes

Concrete sewer pipe 250 mm and smaller shall in general be in accordance with ASTM C14 non-reinforced, socket and spigot or tongue and groove sewer pipe. If trafficability is an issue or for another reason reinforced pipes are to be used they shall comply with ASTM C76, reinforced sewer pipe. The class of the pipes shall be as specified on the drawings.

Concrete sewer pipe 300 mm and larger shall be in accordance with ASTM C76, reinforced sewer pipe. The class and wall thickness shall be as specified on the drawings.

Joints shall be sealed using elastomeric "O" rings in accordance with ASTM C443 unless otherwise specified on the drawings.

4.2.4 Cast/Ductile Iron Soil Pipe

These pipes and fittings shall be hub and plain end coated service weight in accordance with ASTM A74.

Joints shall be neoprene sleeve gasket, push-on type in accordance with ASTM C564.

4.2.5 Concrete Drainage Pipes

All concrete culvert pipes shall be reinforced concrete drainage pipes and shall comply with ASTM C1417.

4.2.6 Corrugated Polyethylene Pipe (CPP)

CPP and fittings shall be socket and spigot type, manufactured in accordance with AASHTO M 294. Socket shall be an integral part of the pipe.

Pipe shall be Type S with a smooth inner liner unless otherwise specified on the drawings.

Joints shall be sealed using elastomeric gaskets in accordance with ASTM F477 unless otherwise specified on the drawings.

4.2.7 Corrugated Steel Drainage Pipes

Pipes shall be in accordance with ASTM A796/A796M for Helical Lockseam Corrugated Steel Pipes and ASTM A760/A760M for Corrugated Steel Pipes, Pipe Arches and Arches, and shall be of the classes and sizes shown on the Drawings. Corrugated steel pipes bent, distorted, badly marked or with surface damage shall be rejected.

4.2.8 Concrete Box Culverts

All concrete box culverts shall be constructed from reinforced concrete in accordance with ASTM C1433.

The appropriate table of section requirements from the listed ASTM standard shall be used for the design loading specified on the drawings.

Joints shall be pre-formed plastic gasket-type joint sealant, unless specified otherwise on the drawings.

4.2.9 Vitrified Clay Pipe (VCP)

Vitrified Clay pipes shall be used only to the extent required to tie into existing VCP systems. For the new piping installations alternative materials should be selected.

Vitrified Clay Pipes shall be extra strength in accordance with ASTM C700.

Joints shall be compression type in accordance with ASTM C425, unless otherwise specified on the drawings.

4.2.10 Covers and Frames for Manholes, Gullies and Catchpits

Covers and frames for manholes, gullies and catchpits shall be precast (monolithic).

Precast concrete manholes shall conform to ASTM C478.

HDPE plate used for manholes and inlet structures shall be manufactured from virgin high-density polyethylene resin in accordance with ASTM D1248, IIIC5-P34 pipe grade resin.

HDPE pipe components used for manholes and inlet structures shall be as specified in Section 4.2.2.

Site-built manholes shall be in accordance with the drawings.

4.3 Backfill Materials

Backfill materials shall conform to the requirements of Ma'aden specification MD-101-SMEM-EG-CV-SPC-0001 – Earthworks.

4.4 Rip Rap

Stones for rip rap shall be sound and free from excessive weathering and laminations. The material shall be durable to withstand long exposure to weathering, and shall not be subject to erosion or disintegration in water. The stones shall be hard and dense and of angular and irregular shape, having a minimum dimension not less than two thirds of the maximum dimension.

Unless otherwise specified, the rip rap shall have grades sizes in accordance with the following:

Table 4-2

Nominal Size	Distribution	% Passing
400	400 mm 300 mm 200 mm 100 mm	100 85 50 15
200	200 mm 150 mm 100 mm 50 mm	100 85 50 15

5. SITE WORKS

5.1 Set Out and Inspection

5.1.1 General

The Contractor shall ensure that all the earthworks are complete prior to proceeding with the installation of storm water drainage and gravity sewers.

The Contractor shall adhere to the drawings. No alterations shall be allowed unless written approval is sought from the Employer.

Should the Employer require to check the set out of any work, the Contractor shall render all the necessary assistance including the provision of any labor and equipment that may be required.

5.1.2 Sewer and Drainage Lines and Encasement

The Contractor shall set out the sewer and drainage lines by the use of overhead profile strings, sighting boards or laser alignment equipment.

Each sewer and drainage line shall be surveyed and a survey record submitted to the Employer. The record shall outline all reference pegs, offset pegs, reduced levels of dumpy level pegs, distances, height of the boning rods and any other relevant information.

The lengths of sewer and drainage lines on drawings may be scaled distances. In the event that an actual measured distance varies greater than 1.5 m from that shown on the drawings in any situation, the Contractor shall notify the Employer and obtain approval prior to proceeding with the Work.

When pegging manholes, the licensed surveyor shall provide at least two reference stakes at 5 m offsets on either side of the manhole centerline and ensure that the level is in accordance with the drawings.

Where required in the drawings, the sewer shall be encased in concrete. In addition, where the length of the sewer exceeds a length of 12 m, an expansion joint shall be constructed within the encasement at 12 m intervals. The joint shall be 15 mm wide and filled with fiber board or an approved filling.

5.2 Survey

The Contractor shall employ a licensed surveyor to set out the drainage and sewerage system in accordance with the drawings. The licensed surveyor shall install survey pegs outlining the existing services, other constraints, road locations and boundary limits, where applicable.

A licensed surveyor shall carry out the following survey works:

- Setting out the works
- 'As constructed' survey. All drawings and information from the 'As Constructed' survey shall be provided to the Employer in an approved format.

5.3 Alignment and Facilities

All lines and levels shown on the drawings are subject to confirmation on Site. The Contractor shall be solely responsible for taking all measures necessary to ensure the accurate location of new work, and its alignment with existing work. The Employer will supply benchmarks.

5.4 Temporary Works

The Contractor shall be responsible for the safe and efficient conduct of traffic through and around the Site of Work at all times.

By-pass roads, detours and other temporary works, including all warning lights, signs, fences and the like, that may be necessary for the safe and orderly deviation of traffic around the works, shall be constructed and maintained. Details of the proposed temporary works shall be provided to the Employer for approval prior to the start of those works.

All warning signs and devices shall be constructed and positioned in accordance with relevant Standards.

5.5 Existing Services

The Contractor shall ascertain the position of and be responsible for the protection of all services in the vicinity of the works, by Site inspection and by contacting the relevant service authorities.

The Contractor shall take every precaution to ensure that existing services are not damaged by its operations before commencing works near services.

The Contractor shall be responsible for the cost and repair of any identified services damaged during the execution of the Work.

Any services not shown on the drawings or identified in the permit to dig and encountered during the progress of the works shall be brought to the attention of the Employer immediately and shall be marked on the 'As Constructed' drawings.

The level of all existing sewer inspection openings, valve pits, service pit lids and similar surface fixtures shall be adjusted to suit the finished ground or pavement level adjacent to the surface fixtures.

5.6 Excavation

5.6.1 Trench Excavations

All excavations shall be performed in accordance with this specification and Ma'aden Specification MD-101-SMEM-EG-CV-SPC-0001 – Earthworks. A Permit to Dig shall be obtained from the Employer prior to commencement of any excavation works. The Contractor shall not commence excavations until after the pegging has been completed.

The line of cut for each side of the trench shall be marked out on the ground surface prior to excavation commencing.

The trenches shall be in accordance with the drawings. The trench shall be kept free from water and any loose debris or disturbed material.

The width of the trenches shall be kept to a minimum, subject to widths required for compaction of backfill.

Trenches shall be dug in advance of either the culvert or pipe laying and be sufficiently long enough to enable works to proceed.

5.6.2 Limits of Excavation

Not more than 100 m of trench shall be excavated ahead of pipe laying without the Employer's approval. However, when weather, soil conditions or other Site restrictions are unfavorable, the Employer may further limit the length of trench that may be excavated ahead of pipe laying.

5.6.3 Manhole and Pit Excavations

The excavations for manholes, pits and storm drains shall be in accordance with the dimensions shown on the drawings. The Contractor shall allow sufficient additional space to carry out the works. All excavations shall be protected with bunting and guard rails and protected at all times while open.

5.6.4 Excavations in Roadways

Where excavations are to be carried out on land that is controlled by a Local Authority, the Contractor shall notify the Authority and seek their approval prior to commencing excavations. In addition the Contractor shall provide traffic control when necessary and keep the excavation to the minimum necessary to complete the works.

5.6.5 Over Excavation

Excavation outside the works limits shall be replaced with approved backfill compacted to an equivalent density of the adjacent undisturbed material, or low strength lean concrete or 5% cement stabilized sand infill at the Contractor's expense.

5.6.6 Barricading of Excavations

All open excavations shall be barricaded with solid barriers or as approved by the Employer.

5.6.7 Drainage of Excavations

As far as is practical the Contractor shall construct excavations to be self-draining and provide drains or bunds around excavations to ensure that run-off does not enter. In the event that water does collect in excavations, provision shall be made to pump them dry. Drainage shall be arranged so as not to damage adjacent property, works, and the environment.

In the event of water being encountered, the trench shall be dewatered and kept clear of water until all bedding is placed and compacted and the pipe, culvert, etc has been placed and backfilled sufficiently to prevent uplift.

5.6.8 Temporary Support (Shoring)

The Contractor shall be responsible for the design and installation of all temporary support required to maintain the excavated surface in its original position and to prevent collapse of the excavation.

Remedial works required resulting from insufficient temporary support shall be the Contractor's responsibility and this works shall be at the Contractor's expense.

Where timbering or sheet piling is used, the specified dimensions of the trench shall be to the inside of the timbering or sheet piling.

Unless otherwise directed by the Employer, timbering and sheet piling shall be removed as backfilling proceeds.

5.6.9 Inspection of Excavation

Before any bedding is placed, the trench bottom shall be inspected by the Employer, who may order any soft or unsuitable patches to be dug out and replaced with approved fine crushed rock or other approved material compacted to an equivalent density of the adjacent undisturbed material.

5.6.10 Insecure Trench Bed

Where the trench bed is discovered to be insecure, or as specified on the drawings, timber or concrete keels in conjunction with timber piles, if required, are to be used if approved by the Employer.

The Contractor shall obtain written approval from the Employer prior to installing any type of keel. The Employer shall inspect and advise whether the trench bed is insecure.

Concrete keels shall be made by using a 1-part cement to 1-part fine aggregate to 2-parts coarse aggregate by volume mix or by using a 20 MPa characteristic compressive strength pre-mix concrete. The size of the keels shall be in accordance with the drawings.

Timber keels shall consist of 200 mm x 50 mm bearers and 200 mm x 38 mm decking of approved timber unless otherwise specified on the drawings.

Timber piles shall consist of 100 mm x 100 mm sections of approved timber and shall be placed at no more than 1 m centers (unless specified otherwise on the drawings) to support the entire length of the keel. Each pile shall be driven into undisturbed ground until the maximum penetration from 14 blows of a 32 kg dolly free falling 1.0 m does not exceed 25 mm.

5.7 Compaction and Moisture Content

The founding base (except non-cohesive material) to which any service is to be placed upon shall be conditioned to obtain characteristic moisture content of 0% – 2% of Optimum Moisture Content and compacted to achieve a minimum dry density of 95% of the maximum Standard Proctor density, as determined by ASTM D698.

The Employer shall approve the foundation pad prior to proceeding with the works.

5.8 Backfilling

5.8.1 General

No backfilling shall take place until all the pipework has been inspected, tested, surveyed and approved by the Employer.

Backfilling shall be in accordance with Ma'aden Specification MD-101-SMEM-EG-CV-SPC-0001 – Earthworks, with the exception of the items outlined in this specification.

5.8.2 Backfill Trenches

5.8.2.1 Non-Trafficable Areas

Pipes located in trenches that are situated in non-trafficable areas shall be backfilled as follows:

- Sand fill shall be hand placed either dry or fully wet and tamped around the barrel from both sides simultaneously to avoid lateral movement of the pipe
- Further similar sand shall be placed above the pipe and compacted in accordance with ASTM D2321 – 05 Clause 7.5
- The remainder of the trench shall be backfilled by mechanical means in layer thicknesses that suit the backfill material and compaction method. The backfill shall be compacted to achieve a minimum dry density of 95% of the maximum Standard Proctor density, as determined by ASTM D698 or compacted to the same density as that of the surrounding undisturbed ground.
- Where applicable the final layer shall be the original topsoil.

5.8.2.2 Trafficable Areas and Encased Pipes

Pipes located in trenches that are situated in trafficable areas and have been concrete encased shall be backfilled as follows, unless specified otherwise on the drawings:

- Cement stabilised sand with a ratio of 100 kg of cement to 1 m^3 of sand shall be placed in layers of thickness not exceeding 225 mm to a height of 300 mm above the pipe. Water shall be added to the fill such that the moisture content is just sufficient to enable mixing and placement of the material. Each layer shall be compacted with a vibratory plate compactor having a minimum static mass of 50 kg.
- A minimum of four passes per layer is required.

- Following the placement of the cement-stabilised sand, a cement-stabilised 'backfill material' shall then be placed in layers of thickness not exceeding 225 mm to within 200 mm of the ground level. The cement-stabilised backfill material ratio shall be 100 kg of cement to 1 m^3 of backfill material. Water shall be added to the fill such that the moisture content is just sufficient to enable mixing and placement of the material. Each layer shall be compacted with a vibratory plate compactor having a minimum static mass of 50 kg.
- A minimum of 6 passes per layer is required
- The final 200 mm depth shall be filled with an approved graded crushed rock base course material, placed and compacted at optimum moisture content to 98% of the modified maximum dry density of the material.

5.9 Rip Rap Placement

5.9.1 General

Rip rap shall be placed in the locations and to the thicknesses shown on the drawings, or as directed by the Employer, and shall be machine or hand placed and not dumped. All rip rap shall be well graded within the sizes specified, and shall so far as is practicable be placed with the smaller stone against the earthworks to be protected so as to prevent loss by scour throughout the interstices.

5.9.2 Rip Rap Placement

Rip rap shall be placed to produce a well graded mass of rock with a minimum percentage of voids.

The larger stones shall be well distributed and the entire mass of stones in their final position shall be roughly graded to conform to the profiles specified. The finished rip rap shall be free from pockets of small stones and clusters of larger stones.

Rip rap shall not be placed in layers or by dumping into chutes or by similar methods likely to cause segregation of the various sizes.

Re-arranging of individual stones by mechanical equipment or by hand shall be carried out if necessary to obtain a reasonably well graded distribution of stone sizes as specified.

5.10 Reinstatement of Roads

After backfilling the trench to the underside of the base course, the base course shall be cut back 200 mm wider than the trench (as shown below) and reinstated using crushed rock to the requirements of Section 5.8.2.2. The surfacing shall then be reinstated to match the type and levels of the existing surface treatment, to the approval of the Local Authorities and/or the Employer.

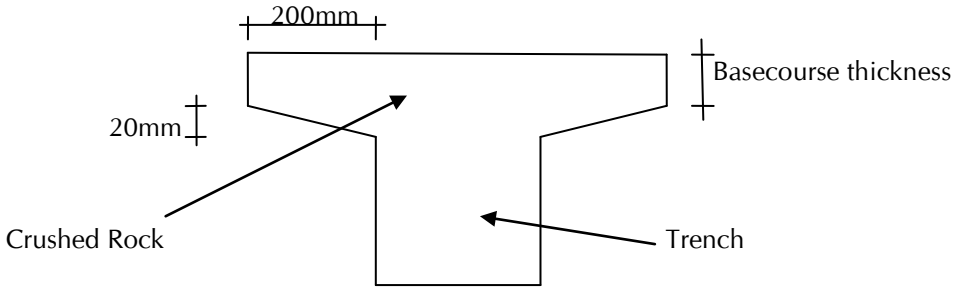


Figure 5-1

5.11 Disposal of Soil

Good quality material excavated from trenches and/or tunnels shall be used for backfilling with Employer's approval. Surplus material shall be disposed of in an approved manner to spoil, stockpile, road or other filling. Soil temporarily stockpiled on the site shall be loaded and transported when required for backfilling of the trenches and/or tunnels.

5.12 Clean-Up

At the completion of the works the surrounding site shall be reinstated to the original condition, and the area shall be cleaned and left in a tidy manner. The removal of spoil and debris shall be the responsibility of the Contractor. The area shall be left in a clean and tidy state to the approval of the Employer and/or the Local Authorities.

In addition the Contractor shall ensure that the inside of the sewerage and drainage system is left clean by flushing the system with water.

6. STORM WATER DRAINAGE

6.1 General

The drainage system shall be installed in accordance with this specification and the drawings.

The Contractor shall ensure that prior to and during the works that all the necessary safety precautions shall be taken with regard to people, equipment, existing services, nearby buildings, structures, etc. In the event of damage the Contractor shall make good at the Contractor's expense to the approval of the Employer and or Authorities concerned.

6.2 Precast Concrete Box and Pipe Culverts

6.2.1 Box and Pipe Culvert Bedding

The Contractor shall place the Precast Box Culvert bases, or construct a cast in-situ base slab to the requirements of the drawings, on a continuous, suitably leveled bedding material evenly compacted to the requirements of Section 5.7.

The bedding material shall be as specified in the contract documents and shall be of a dry and non-cohesive nature. The Employer shall approve the bedding prior to placement.

6.2.2 Installation of Culverts

All pipes and culverts shall be laid in accordance with the manufacturer's recommendations. Under no circumstances shall blocking be used to support the base of either pipes or box culverts in-situ.

The Contractor shall be responsible for providing the lifting aids for lowering the sections into the trenches. Box culvert sections shall be handled and placed using the cast in lifting points. Slings shall not be used.

The Contractor shall ensure that the culverts are laid to the grades and alignments indicated on the drawings.

6.2.3 Concrete Pipe Culverts

All concrete pipe culverts greater than 600 mm in diameter shall have external or internal flush joints. Pipes having a diameter of 600 mm or less shall have socket and spigot type joints.

The Contractor shall ensure that when laying the pipe and fittings that the socket end is placed facing upstream. Prior to fitting the pipes together, the Contractor shall ensure that all dirt, dust and foreign matter is removed from the pipe joint.

In addition, the neoprene rubber ring shall be placed on the spigot of the pipe to be laid and then fitted into the socket of the pipe previously laid in the trench. Care shall be taken to ensure complete sealing, prevention of distortion to the ring and minimal disturbance of the bedding material.

6.2.4 Concrete Box Culverts

Immediately prior to either placing the crown on the base slab (legs down case), or the lid on the inverted box culvert (legs up case), a pad of suitable grout shall be applied to the mating surfaces such that they are fully and evenly supported. For culverts that have a span greater than 1200 mm, the gap between the inside of the leg and the recess shall be grouted.

The Contractor shall ensure that the box culverts and the lids or pre-cast bases are not staggered but placed to achieve vertical alignment in all directions.

When placing backfill the Contractor shall ensure that the fill placement does not damage any section of culvert or unevenly load the culverts.

6.2.5 Pits and Headwalls

Entry and junction pits and headwalls shall be made from reinforced concrete in accordance with the relevant standards.

The drainage pits shall be supplied with gratings, covers and step irons in accordance with the drawings and relevant standards.

6.2.6 Culvert and Open Plan Outlet Rock Protection

Culvert outlet rock protection shall be constructed in accordance with the drawings and/or as directed by the Employer.

Materials used for culvert outlet protection shall comply with Section 4.4.

6.3 Open Earth Drains

Open earth drains shall be constructed in accordance with the drawings and/or as directed by the Employer.

The Contractor shall ensure that the drains are constructed with a smooth gradient with no obstructions or hollows that could impede the flow of water or allow ponding.

Where shown on the drawings, the drains shall be protected from erosion by constructing an apron, comprised of rock not less than 100 mm thick in 1:3 cement: sand mortar mix. The Contractor shall ensure that all rocks are roughly the same size with a surface area of approximately 0.03 m^2.

6.4 Tolerances

6.4.1 Tolerance for Pipe laying

The pipe lengths shall be constructed to alignment, levels and gradients as defined on the Drawings. The allowable tolerances from such lines and levels shall be as follows:

- 25 mm from the specified alignments
- 5 mm from the specified invert levels provided that the deviation from the specified gradient of the drainage line is not greater than 0.1%.

6.4.2 Tolerance for Service Pit Covers and Grates

The allowable tolerances for the top of manhole covers and pits shall be as follows:

- +0, – 5 mm for tops with grates for drainage purposes
- -0, +5 mm for tops in roads and pavements where grates are not provided
- +20, +50 mm above the finished surface level other than in roads and pavements for tops where grates are not provided.

6.4.3 Tolerance for Rip Rap

The allowable tolerances for rip rap placement shall be $\pm$100 mm from the defined levels and grades provided that the rip rap material is not higher than the outlet invert level of adjacent storm water pipes and culverts.

7. SEWERAGE

7.1 Pipe Bedding

Sewerage pipes shall be placed on bedding in accordance with the drawings.

The type of bedding material used in the trenching shall be dependent on the worst condition encountered in the trench between two adjacent manholes. Only one type of bedding material shall be used for a sewer line between two adjacent manholes.

The bedding shall be placed such that a smooth gradient is achieved in accordance with the drawings for the entire length of the pipe and it shall be compacted as per Section 5.7.

Where the pipe is in a trench excavated in rock, bedding shall consist of sand or sand/gravel minimum 300 mm thick unless otherwise indicated on drawings.

Where the pipe is to be laid in rock, the bedding shall consist of crushed aggregate to a minimum depth of 150 mm unless specified otherwise on the drawings.

7.2 Pipe Laying

Pipe laying and jointing shall not be carried out unless the Employer has approved the condition of the trench bottom and the type of bedding, surround or foundation.

In all cases, pipes shall be bedded evenly for the full length of the barrels with sockets free from pressure. Where flexible joints are used, the pipes shall be free to move along their lengths and shall not be bonded to concrete bedding or surround.

Unless specifically required on the drawings, pipes shall be laid in straight lines between service pits with sockets facing up grade. Spigots and sockets shall be cleaned and the interiors of pipes shall be free from obstructions.

Pipes manufactured with elliptical reinforcement shall be laid top up.

Where pipes are inserted into sleeves, the pipe shall be cut mechanically so that the cross sectional surface remains smooth, clean and flush. The joint between pipe and maintenance structure (i.e. manhole etc.) shall be mortared comprising one part cement by volume to three parts of sand.

The Contractor shall ensure that pipes and joints are free of leakage after laying. This may require verification by suitable test methods as agreed with the Employer.

7.3 Pipe Joints

Pipe jointing shall conform to Section 4.2.

7.4 Manholes Construction

7.4.1 General

Manholes shall conform to Section 4.2.10.

The manholes shall have sealed inlets, which consist of a 300 mm length of pipe that is of the same material and size as that proposed for the sewer line feeding into the manhole. The inlet pipe shall be cast into the manhole at the correct invert level and gradient.

The Contractor shall ensure that the position of the manhole is in accordance with the drawings, and that no variance from that shown on the drawings shall occur unless written approval is obtained from the Employer.

7.4.2 Brick Manholes

Brick manholes shall be constructed in accordance with the drawings and this specification. Brick manholes shall be classified as either 'wet' or 'dry' type.

The bricks are to be soaked in water and clean from any grit prior to being used and shall be installed with neat joints not exceeding 10 mm and plumbed and true to line.

Wet type manholes are to be used when the known maximum water table level is higher than 600 mm from the lowest invert level of the manhole. For wet type manholes, the brick cavity shall be filled with cement mortar.

7.4.3 Cast-in-situ Concrete Manholes

Cast-in-situ concrete manholes shall be constructed in accordance with the drawings and this specification.

Cast in situ concrete used for manholes shall be constructed in accordance with MD-101-SMEM-EG-ST-SPC-0005 – Concrete Works.

The Contractor shall ensure that the walls have a smooth finish with no defects.

7.4.4 Pre-Cast Concrete Manholes

Pre-cast concrete manholes shall be installed in accordance with the drawings and the manufacturer's instructions. Only approved manufacturers shall be used.

7.4.5 Manhole Bases

Manhole bases shall be constructed in concrete in accordance with the drawings and this specification. The bases shall be constructed on sound compacted foundations, with the exception of when the ground encountered is insecure and in such case the base shall be constructed on piles, bearers and decking.

Benching shall be placed in accordance with the drawings. The minimum distance between the invert of channels and the manhole base shall be 70 mm. Channels and benching shall have steel trowelled finished in accordance with the relevant Standards.

7.4.6 Manhole Covers

Manhole covers shall conform to Section 4.2.10.

For trafficable areas the manhole covers shall be of a cast iron frame type including a cast iron support frame. For non-trafficable areas concrete type covers will be allowed.

For manholes located in roadways the cover shall finish flush with the surrounding finished surface level.

For manholes located in virgin soil, the cover shall finish 50 mm above the surrounding ground level and sloped to suit.

7.4.7 Manhole Ladders

For manholes having a depth of 3.5 m or less, stainless steel ladder hooks shall be installed on the upstream wall of the manhole.

For manholes having a depth of more than 3.5 m, a mild steel hot dipped galvanized ladder shall be installed on the upstream wall of the manhole.

For brick manholes where the sewers are deeper than 1.8 m, mild steel hot dipped galvanized step irons shall be used. Step irons shall be galvanized in accordance with the relevant Standards.

7.5 Level Tolerances

When installing the sewer lines the following tolerances shall be adhered to.

7.5.1 Horizontal Tolerances

- No greater than 150 mm from the position outlined on the drawings
- No greater than 5 mm in 1m of the pipeline from the straight line
- No greater than 25 mm from the straight line between the centreline of adjacent manholes.

7.5.2 Vertical Tolerances

- The pipeline shall have a continuously falling gradient along the entire length
- No greater than 10 mm deviation of the invert level from that shown on the drawings.

8. AS BUILT DRAWINGS

Prior to backfilling the Contractor shall survey the entire system via the use of a licensed surveyor.

The survey record shall include:

- All invert levels at manholes and inspection openings
- Distances from the centres of manholes/pits to adjacent manholes pits
- Location of manholes/pits in relation to survey control points
- Reduced levels of manhole covers
- Any other relevant information.

Underground services drawing shall be updated. The survey records shall be used to produce as-built drawings.

9. TESTING

The Contractor shall submit for the Employer's approval an Inspection and Test Plan (ITP) which shall outline tests to be carried out to certify construction to the requirements of the specification, including hold points. As a minimum requirement, hold points shall be noted:

- Prior to laying of pipes or conduit and manholes, and
- Prior to backfilling.

The Contractor shall provide all test equipment and carry out all testing in accordance with the approved ITP to show that a particular section of the works has been constructed in accordance with the relevant standards. Copies of all tests shall be upon request by the Employer.

Reworking of a failed section shall continue until the section complies with the specification.

The Contractor shall engage an independent registered sub-contractor for the purpose of carrying out laboratory and field-testing of concrete, soils and aggregates.

The Contractor shall conduct in-situ density tests to confirm specified compaction requirements have been achieved. The Employer may witness such tests and the results shall be made available to the Employer upon request.

The Contractor shall obtain the necessary documentation from the manufacturer confirming that the coating of pipes is completed as per the relevant standards. The documentation shall be submitted to the Employer upon request.

The Contractor shall inspect the coating of the manholes as applicable, and shall randomly inspect the coating of the pipes on Site to confirm that there is no damage in transportation. In case of a damage, the pipe spool shall be repaired or replaced.

10. QUALITY CONTROL

The Contractor shall refer to the following.

MD-101-SMPM-PM-QA-PEI-0001 Contractor Quality Requirements